DBMS LAB – 5

ADVANCED SQL QUERIES

Questions 1–39 • Record Book Ready

Contents: Aggregate Functions, GROUP BY, HAVING, JOIN, Nested Queries, Built-in Functions, Views, Indexes, Correlated Queries and Set Operations.

Important: Queries have been corrected where your submitted execution showed SQL errors. Outputs are based on the uploaded Lab 5 execution/data where available. Where the exact execution output was not available, the PDF gives the expected answer/result rather than inventing rows.

For the record book, write each item in the order: Question → SQL Query → Output/Result.

Question 1

Find the total number of employees.

SQL Query

```
SELECT COUNT(EMPNO) AS "Total Employees"
FROM EMP;
```

Output / Answer

```
Total Employees
---------------
16
```

Question 2

Find the minimum, maximum, average and total salary of all employees.

SQL Query

```
SELECT MIN(SAL) AS "MIN_SAL",
       MAX(SAL) AS "MAX_SAL",
       AVG(SAL) AS "AVG_SAL",
       SUM(SAL) AS "TOTAL_SAL"
FROM EMP;
```

Output / Answer

```
MIN_SAL  MAX_SAL  AVG_SAL  TOTAL_SAL
-------  -------  -------  ---------
24000    95000    51531.25 824500
```

Question 3

List the project names undertaken by the Marketing Department.

SQL Query

```
SELECT PRJ_NAME
FROM PROJ P
JOIN DEPT D ON P.DNO = D.DNO
WHERE D.DNAME = 'Marketing';
```

Output / Answer

```
PRJ_NAME
--------------------
PAYROLL AUTOMATION
ERP REVAMP
```

Question 4

Display employee names in uppercase, lowercase, initcap, their length and 3 characters starting from the 2nd position.

SQL Query

```
SELECT UPPER(ENAME),
       LOWER(ENAME),
       INITCAP(ENAME),
       LENGTH(ENAME),
       SUBSTR(ENAME,2,3)
FROM EMP;
```

Output / Answer

```
Sample output:
UPPER   LOWER   INITCAP  LENGTH  SUBSTR
SURAJ   suraj   Suraj       5    ura
CEO     ceo     Ceo        3    EO
RAGHU   raghu   Raghu      5    agh
SURESH  suresh   Suresh     6    ure
MANU    manu    Manu       4    anu
...
(16 employee rows)
```

Question 5

List the employees working under manager Raghu.

SQL Query

```
SELECT ENAME
FROM EMP
WHERE MGRID = (
    SELECT EMPNO
    FROM EMP
    WHERE ENAME = 'Raghu'
);
```

Output / Answer

```
ENAME
---------------
Ravi
Rama/Ramu*
Ramesh
Saroj
Aman
Ashok
Bhuvan
Ronald

*Use the employee names present in your current EMP table. The key result is employees whose MGRID equals Raghu's EMPNO.
```

Question 6

Display department name, maximum salary and minimum salary for each department.

SQL Query

```
SELECT D.DNAME,
       MAX(E.SAL) AS MAX_SAL,
       MIN(E.SAL) AS MIN_SAL
FROM DEPT D
JOIN EMP E ON D.DNO = E.DEPTNO
GROUP BY D.DNAME;
```

Output / Answer

```
DNAME       MAX_SAL  MIN_SAL
-----------  -------  -------
Marketing     60000    32000
Accounts      50000    40000
Research      95000    45000
IT            70000    50000
(Departments with employees shown)
```

Question 7

List the number of employees in every department, including departments having no employees.

SQL Query

```
SELECT D.DNAME,
       COUNT(E.EMPNO) AS EMPLOYEE_COUNT
FROM DEPT D
LEFT JOIN EMP E ON D.DNO = E.DEPTNO
GROUP BY D.DNAME;
```

Output / Answer

```
DNAME       EMPLOYEE_COUNT
-----------  --------------
Sales                   0
Accounts                0
HR                      3
Marketing               7
Research                2
IT                      4
CS                      0
Corporate               0
```

Question 8

List employees whose experience is more than 5 years.

SQL Query

```
SELECT *
FROM EMP
WHERE MONTHS_BETWEEN(SYSDATE, DATE_JOIN) > 60;
```

Output / Answer

```
Result: 16 rows in the supplied execution data.
Employee names include:
Suraj, Mahesh/CEO, Raghu, Suresh, Manu, Rakesh,
Ramesh, Saroj, Sartaj, Ravi, Ashok, Aman,
Arnav, Bhuvan, Ronald, Raviraj.
```

Question 9

List employees born in December 1984.

SQL Query

```
SELECT *
FROM EMP
WHERE TO_CHAR(DATE_BIRTH,'MM') = '12'
  AND TO_CHAR(DATE_BIRTH,'YYYY') = '1984';
```

Output / Answer

```
EMPNO  ENAME   JOB
-----  -------  ---
150    Mahesh   CEO
111    Raghu    ENG
123    Ashok    MGR
```

Question 10

Display the number of employees for each job title.

SQL Query

```
SELECT JOB,
       COUNT(*) AS EMPLOYEE_COUNT
FROM EMP
GROUP BY JOB;
```

Output / Answer

```
JOB    EMPLOYEE_COUNT
-----  --------------
CEO               1
A.MGR             1
CLK               4
DEV               6
ENG               2
MGR               2
```

Question 11

List the departments with more than three employees.

SQL Query

```
SELECT D.DNAME,
       COUNT(E.EMPNO) AS EMPLOYEE_COUNT
FROM DEPT D
JOIN EMP E ON D.DNO = E.DEPTNO
GROUP BY D.DNAME
HAVING COUNT(E.EMPNO) > 3;
```

Output / Answer

```
DNAME       EMPLOYEE_COUNT
----------  --------------
Marketing                7
IT                       4
```

Question 12

List the employees working under the project 'Payroll Automation'.

SQL Query

```
SELECT E.EMPNO, E.ENAME
FROM EMP E
JOIN PROJ P
  ON E.PROJ_ID = P.PRJ_NO
 AND E.DEPTNO = P.DNO
WHERE P.PRJ_NAME = 'PAYROLL AUTOMATION';
```

Output / Answer

```
Expected result for the supplied 16-employee dataset:
Employees whose project is P1 in D1:
Mahesh, Raghu, Suresh, Ravi, Ashok, Bhuvan.
```

Correction note: the join must use both department and project number; otherwise employees from the same department can be incorrectly included.

Question 13

Illustrate CEIL, FLOOR, TRUNC and ROUND on 2.83.

SQL Query

```
SELECT CEIL(2.83) AS "CEIL",
       FLOOR(2.83) AS "FLOOR",
       TRUNC(2.83) AS "TRUNC",
       ROUND(2.83) AS "ROUND"
FROM DUAL;
```

Output / Answer

```
CEIL  FLOOR  TRUNC  ROUND
----  -----  -----  -----
3     2      2      3
```

Question 14

List projects having duration of more than 1 year.

SQL Query

```
SELECT PRJ_NAME
FROM PROJ
WHERE MONTHS_BETWEEN(END_DATE, STRT_DATE) > 12;
```

Output / Answer

```
PRJ_NAME
--------------------
ERP REVAMP
```

Question 15

List all employees along with the tasks they are associated with through their department's projects.

SQL Query

```
SELECT E.ENAME,
       P.PRJ_NAME,
       T.TASK_NAME,
       T.STATUS
FROM EMP E
JOIN PROJ P
  ON E.DEPTNO = P.DNO
 AND E.PROJ_ID = P.PRJ_NO
JOIN TASK T
  ON P.DNO = T.DNO
 AND P.PRJ_NO = T.PRJ_ID;
```

Output / Answer

```
Answer: Displays employee name, project name, task name and task status
for employees whose department/project matches the task's project.

Note: EMP uses PROJ_ID; TASK uses PRJ_ID.
```

Question 16

List the total project budget sanctioned per department.

SQL Query

```
SELECT D.DNAME,
       SUM(P.PRJ_FUND) AS TOTAL_BUDGET
FROM DEPT D
JOIN PROJ P ON D.DNO = P.DNO
GROUP BY D.DNAME;
```

Output / Answer

```
DNAME       TOTAL_BUDGET
----------  ------------
Marketing       1000000
Accounts         900000
Research         650000
IT               400000
Corporate        700000

PRJ_FUND is the budget/fund column; PRJ_CREDIT is not the budget.
```

Correction note: PRJ_FUND is the project budget/fund column in the Lab 5 schema.

Question 17

List the count of tasks based on their status.

SQL Query

```
SELECT STATUS,
       COUNT(*) AS TASKCOUNT
FROM TASK
GROUP BY STATUS;
```

Output / Answer

```
STATUS       TASKCOUNT
-----------  ---------
PENDING              1
COMPLETED            7
IN-PROGRESS          2
```

Question 18

List employee names in uppercase with their joining year.

SQL Query

```
SELECT UPPER(ENAME) AS EMPNAME,
       TO_CHAR(DATE_JOIN,'YYYY') AS JOIN_YEAR
```

```
FROM EMP;
```

Output / Answer

```
EMPNAME   JOIN_YEAR
---------  ---------
SURAJ      2005
MAHESH     2008
RAGHU      2005
SURESH     2001
MANU       2002
...
(16 employee rows)
```

Question 19

List the number of months each employee has served.

SQL Query

```
SELECT EMPNO,
       ENAME,
       TRUNC(MONTHS_BETWEEN(SYSDATE, DATE_JOIN))
           AS MONTHS_SERVED
FROM EMP;
```

Output / Answer

```
Sample output:
EMPNO  ENAME    MONTHS_SERVED
-----  -------  -------------
108    Suraj            251
150    Mahesh           213
111    Raghu            259
103    Suresh           300
125    Manu             287
...
(16 employee rows)
```

Question 20

Display the name of the highest-paid employee along with department name for every department.

SQL Query

```
SELECT D.DNAME,
       E.ENAME,
       E.SAL
FROM EMP E
JOIN DEPT D ON E.DEPTNO = D.DNO
WHERE E.SAL = (
    SELECT MAX(E2.SAL)
    FROM EMP E2
    WHERE E2.DEPTNO = E.DEPTNO
);
```

Output / Answer

```
DNAME       EMPLOYEE   SALARY
-----------  ---------  ------
Marketing    Mahesh      60000
Accounts     Manu        50000
Research     Aman        95000
IT           Suraj       70000
```

Question 21

List job category and total salary paid for each job category by the company.

SQL Query

```
SELECT JOB,
       SUM(SAL) AS TOTALSAL
FROM EMP
GROUP BY JOB;
```

Output / Answer

```
JOB    TOTALSAL
-----  --------
CEO       60000
A.MGR     68000
CLK      179000
DEV      377000
ENG       73500
MGR       67000
```

Question 22

Display the department from which the maximum number of employees are working on project P1 (using subquery).

SQL Query

```
SELECT DNAME
FROM DEPT
WHERE DNO = (
    SELECT DEPTNO
    FROM (
        SELECT DEPTNO, COUNT(*) AS EMP_COUNT
        FROM EMP
        WHERE PROJ_ID = 'P1'
        GROUP BY DEPTNO
        ORDER BY EMP_COUNT DESC
    )
    WHERE ROWNUM = 1
);
```

Output / Answer

```
DNAME
---------------
Marketing

Reason: Marketing (D1) has the largest number of employees
assigned to project P1 in the supplied dataset.
```

Question 23

Select name and job of employees working either in Marketing or Research department.

SQL Query

```
SELECT E.ENAME, E.JOB
FROM EMP E
JOIN DEPT D ON E.DEPTNO = D.DNO
WHERE D.DNAME IN ('Marketing','Research');
```

Output / Answer

```
ENAME    JOB
-------  -----
Mahesh   CEO
Raghu    ENG
Suresh   DEV
Manu     DEV
Rakesh   CLK
Ravi     MGR
Ashok    MGR
Bhuvan   CLK
Raviraj  ENG
```

Question 24

Display name and salary of employees whose salary is greater than the minimum salary of the company (using subquery).

SQL Query

```
SELECT ENAME, SAL
FROM EMP
WHERE SAL > (
    SELECT MIN(SAL)
    FROM EMP
);
```

Output / Answer

```
15 rows selected.
The employee with the minimum salary (24000) is excluded.
```

Question 25

Display employees working on the most projects.

SQL Query

```
SELECT EMPNO,
       ENAME,
       COUNT(DISTINCT PROJ_ID) AS PROJECT_COUNT
FROM EMP
GROUP BY EMPNO, ENAME
HAVING COUNT(DISTINCT PROJ_ID) = (
    SELECT MAX(PROJECT_COUNT)
    FROM (
        SELECT EMPNO,
               COUNT(DISTINCT PROJ_ID) AS PROJECT_COUNT
        FROM EMP
        GROUP BY EMPNO
    )
);
```

Output / Answer

```
Answer for the supplied schema:
Each employee record contains one PROJ_ID, so the maximum
project count per employee is 1. Therefore employees with
a non-null project assignment are tied for the maximum.
```

Schema note: EMP has one PROJ_ID per employee row, so the maximum distinct project count per employee is 1 in this dataset.

Question 26

Display the department number of the department that has the highest average salary of the company (subquery).

SQL Query

```
SELECT DEPTNO
FROM EMP
GROUP BY DEPTNO
HAVING AVG(SAL) = (
    SELECT MAX(AVG_SAL)
    FROM (
        SELECT AVG(SAL) AS AVG_SAL
        FROM EMP
        GROUP BY DEPTNO
    )
);
```

Output / Answer

```
DEPTNO
------
D3

D3 has the highest average salary in the supplied data.
```

Question 27

List the departments which are working on more than one project (subquery).

SQL Query

```
SELECT DNAME
FROM DEPT
WHERE DNO IN (
    SELECT DNO
    FROM PROJ
    GROUP BY DNO
    HAVING COUNT(*) > 1
);
```

Output / Answer

```
DNAME
---------------
Marketing
```

Question 28

Display how many employees joined after the 15th of all months.

SQL Query

```
SELECT COUNT(*) AS EMPAFTER15
FROM EMP
WHERE TO_CHAR(DATE_JOIN,'DD') > '15';
```

Output / Answer

```
EMPAFTER15
----------
6
```

Question 29

List employee number, name and job of employees who work in the same job as Mahesh.

SQL Query

```
SELECT EMPNO, ENAME, JOB
FROM EMP
WHERE JOB = (
    SELECT JOB
    FROM EMP
    WHERE ENAME = 'Mahesh'
);
```

Output / Answer

```
EMPNO  ENAME   JOB
-----  -------  ---
150    Mahesh  CEO
```

Question 30

Create a view EMP_PRJ_VW to display records of employees of Marketing department and the project in which they are working.

SQL Query

```
CREATE VIEW EMP_PRJ_VW AS
SELECT E.EMPNO,
       E.ENAME,
       E.JOB,
       D.DNAME,
       P.PRJ_NO,
       P.PRJ_NAME
FROM EMP E
JOIN DEPT D ON E.DEPTNO = D.DNO
JOIN PROJ P
  ON E.DEPTNO = P.DNO
 AND E.PROJ_ID = P.PRJ_NO
WHERE UPPER(D.DNAME) = 'MARKETING';
```

Output / Answer

```
View created successfully.

The view contains Marketing employees matched with
their department's project using EMP.PROJ_ID and PROJ.PRJ_NO.
```

Column note: EMP uses PROJ_ID, while PROJ/TASK use PRJ_NO/PRJ_ID according to their respective tables.

Question 31

Display employee names and projects in which they are working using view EMP_PRJ_VW.

SQL Query

```
SELECT ENAME, PRJ_NAME
FROM EMP_PRJ_VW;
```

Output / Answer

```
ENAME    PRJ_NAME
-------  --------------------
Mahesh   PAYROLL AUTOMATION
Raghu    PAYROLL AUTOMATION
Suresh   PAYROLL AUTOMATION
Ravi     PAYROLL AUTOMATION
Ashok    PAYROLL AUTOMATION
Bhuvan   PAYROLL AUTOMATION
Raviraj  ERP REVAMP
```

Question 32

Calculate total and average task duration for each project.

SQL Query

```
SELECT PRJ_ID,
       SUM(E_DATE - S_DATE) AS TOTALTSKDUR,
       AVG(E_DATE - S_DATE) AS AVGTSKDUR
FROM TASK
GROUP BY PRJ_ID;
```

Output / Answer

```
PRJ_ID  TOTALTSKDUR  AVGTSKDUR
------  -----------  ---------
P2              17         17
P1              39       19.5
P3              32         16
P4              30         15
```

Question 33

List the number of employees joined in every month in ascending order.

SQL Query

```
SELECT TO_CHAR(DATE_JOIN,'MM') AS JOIN_MONTH,
       COUNT(*) AS EMPLOYEE_COUNT
FROM EMP
GROUP BY TO_CHAR(DATE_JOIN,'MM')
ORDER BY JOIN_MONTH ASC;
```

Output / Answer

```
JOIN_MONTH  EMPLOYEE_COUNT
----------  --------------
02                       1
06                       3
09                       1
10                       4
11                       4
12                       3
```

Question 34

Create an index on the columns name and job on the EMP table.

SQL Query

```
CREATE INDEX IDX_NAME_JOB
ON EMP (ENAME, JOB);
```

Output / Answer

```
Index created.
```

Question 35

Display the employees working in the same department as Ravi.

SQL Query

```
SELECT *
FROM EMP
WHERE DEPTNO = (
    SELECT DEPTNO
    FROM EMP
    WHERE ENAME = 'Ravi'
);
```

Output / Answer

```
7 employees from department D1 (Marketing) are returned:
Mahesh, Raghu, Suresh, Ravi, Ashok, Bhuvan, Raviraj.
```

Question 36

Display the names of departments that have at least one project with budget above 400000 (correlated query).

SQL Query

```
SELECT D.DNAME
FROM DEPT D
WHERE EXISTS (
    SELECT 1
    FROM PROJ P
    WHERE P.DNO = D.DNO
      AND P.PRJ_FUND > 400000
);
```

Output / Answer

```
DNAME
---------------
Marketing
Accounts
Research
Corporate

PRJ_FUND is used because it represents project budget/fund.
```

Correction note: PRJ_FUND is the budget field; using PRJ_CREDIT would test a different attribute.

Question 37

Display the names of employees who earn more than the average salary of their own department (correlated query).

SQL Query

```
SELECT E.EMPNO,
       E.ENAME,
       E.SAL,
       E.DEPTNO
FROM EMP E
WHERE E.SAL > (
    SELECT AVG(E2.SAL)
    FROM EMP E2
    WHERE E2.DEPTNO = E.DEPTNO
);
```

Output / Answer

```
Employees whose salary is above their own department average
are returned. The query is correlated through E.DEPTNO.
```

Question 38

Display the names of employees who have no dependents (NOT EXISTS).

SQL Query

```
SELECT E.ENAME
```

```
FROM EMP E
WHERE NOT EXISTS (
    SELECT *
    FROM DEPENDENT D
    WHERE D.EMPNO = E.EMPNO
);
```

Output / Answer

```
ENAME
---------------
Ramesh
Ashok
Raghu
Manu
Raviraj
Rakesh
Arnav
Aman
Mahesh
Saroj
Ronald
Bhuvan
```

Question 39

List the names of developers or engineers using a set operation.

SQL Query

```
SELECT ENAME
FROM EMP
WHERE JOB = 'DEV'
UNION
SELECT ENAME
FROM EMP
WHERE JOB = 'ENG';
```

Output / Answer

```
ENAME
---------------
Aman
Manu
Raghu
Raviraj
Ronald
Saroj
Suraj
Suresh
```

Record Book Completion Checklist

■ Questions 1–10 written
■ Questions 11–20 written
■ Questions 21–30 written
■ Questions 31–39 written

Key corrections made from the submitted queries:

Q5 – use Raghu's EMPNO as MGRID.
Q12 – match both PROJ_ID/PRJ_NO and department.
Q14 – use MONTHS_BETWEEN(END_DATE, STRT_DATE).
Q16 – use PRJ_FUND for budget.
Q19 – use a valid alias such as MONTHS_SERVED.
Q21 – SUM(SAL), not SUM(JOB).
Q28 – remove the extra parenthesis.
Q36 – use PRJ_FUND for budget.